

# ONEREPLAY: USING REPLAY DATA WITH CONSTANT COMPLEXITY FOR LLM POST-TRAINING

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## ABSTRACT

This paper presents OneReplayer, a method that transforms a large replay dataset into a fixed-size matrix memory independent of the dataset size, which is then used to mitigate forgetting during LLM post-training. Existing replay-based approaches reduce forgetting by repeatedly revisiting old examples during fine-tuning, but their cost often grows with the accumulated data and becomes especially burdensome when post-training requires many repeated trials for hyperparameter search, data mixture tuning, or alignment refinement. Considering that the historical replay dataset is fixed, we replace example-level replay with a reusable second-order activation memory. This memory constrains functional changes in the updated model without requiring recomputing historical examples at each training step. Experiments across supervised fine-tuning, on-policy distillation, and reinforcement learning consistently demonstrate the effectiveness of our method in preserving post-training task performance while mitigating forgetting.

## 1 INTRODUCTION

Open-source models play a vital role in the advancement of modern AI systems by making frontier AI technologies more accessible and customizable. Users can adapt these models to their specific needs through post-training, thereby broadening their practical applications and contributing to continued progress in AI research. However, a critical challenge is that post-training can lead to severe decrease of the model’s instruction-following capabilities, which are essential to LLM specialists.

Data replay is one of the most effective and widely used strategies for mitigating this issue. By mixing old examples with new post-training data, replay directly constrains the model to preserve previous behavior while acquiring new capabilities. Despite its effectiveness, replay introduces a fundamental cost problem. As the amount of historical data grows, maintaining strong retention often requires revisiting increasingly large or carefully selected replay sets. This cost becomes especially burdensome in frontier LLM research, where training is rarely a single run and the replay set needs to be visited repeatedly. Practitioners often conduct many repeated trials to tune learning rates, data mixtures, LoRA ranks, regularization strengths, alignment objectives, and early stopping criteria. Or even if the training is a single run, the replay data still needs to be used repeatedly due to multi-epoch iterations. If each trial requires repeated access to old replay data, the accumulated cost of replay can become unaffordable for lots of researchers.

A common response is to use a small selected subset of old data (known as replay buffer). Recent work on data-efficient replay, synthetic rehearsal, on-policy replay, and instruction-aware sample selection improves this trade-off by choosing more informative replay examples. However, a small buffer may not necessarily capture the functional constraints imposed by a large and diverse historical dataset, and these methods offer only a partial compromise to the trade-off between efficiency and effectiveness. In this paper, we ask a different question: *can the use of replay data be achieved at a constant cost independent of the dataset size?*

We propose OneReplayer, which stores the replay dataset in a fixed-size matrix. After a one-time scan to convert the dataset into this matrix representation, all subsequent repeat usages are independent of the original dataset size.

## 2 RELATED WORK

## 3 PRELIMINARIES

**LLM Post-Training.** Let  $f_\theta$  denote a pre-trained or previously post-trained large language model parameterized by  $\theta$ . In modern LLM development,  $f_\theta$  is often further adapted to new data through supervised fine-tuning, preference optimization, safety tuning, domain adaptation, or parameter-efficient tuning. We refer to these procedures collectively as post-training.

Given a new post-training dataset  $\mathcal{D}_{\text{new}} = (u_i, y_i)_{i=1}^n$ , where  $u_i$  denotes an input instruction or context and  $y_i$  denotes the target response, standard post-training minimizes

$$\mathcal{L}_{\text{task}}(\theta) = \mathbb{E}_{(u,y) \sim \mathcal{D}_{\text{new}}} [-\log p_\theta(y | u)]. \quad (1)$$

Although post-training improves performance on  $\mathcal{D}_{\text{new}}$ , it may lead to the forgetting of general foundational capabilities, such as instruction-following and safety guardrails, which are prerequisites for performing most specialized tasks.

**Replay for Forgetting Mitigation.** A standard way to address this problem is replay. Instead of training only on  $\mathcal{D}_{\text{new}}$ , replay-based methods optimize a mixed objective

$$\mathcal{L}_{\text{replay}}(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{new}}} [\mathcal{L}(\theta; x, y)] + \beta \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{old}}} [\mathcal{L}(\theta; x, y)], \quad (2)$$

where  $\mathcal{D}_{\text{old}}$  is a replay set or replay buffer, and  $\beta$  controls the strength of old-task preservation. Depending on the setting,  $\mathcal{B}_{\text{old}}$  may contain old instruction-response pairs, safety examples, preference data, etc. Different customized LLMs are designed for different purposes and therefore emphasize different capabilities. Nevertheless, instruction-following is a fundamental capability required for successful task completion in the vast majority of application scenarios. Therefore, this study primarily focuses on the preservation of instruction-following ability during post-training.

Replay is effective because it directly constrains the model’s behavior on old examples during new post-training. However, it also introduces an additional cost. At each training step, the model must access, tokenize, batch, and compute gradients on replay examples. If the amount of historical data grows, strong replay may require either a larger replay set or more careful sample selection. Even when the replay buffer is fixed, its representativeness can deteriorate as the diversity of historical data increases.

This motivates a different question: rather than repeatedly replaying old examples, can we compress the functional constraint induced by old examples into a fixed-size object that can be reused during each training step?

## 4 METHOD

**Objective formulation: behavior retention.** Modern LLMs are largely composed of linear transformations. Consider one such weight matrix  $W_0 \in \mathbb{R}^{m \times d}$  from the initial model checkpoint. During post-training, this matrix is updated to  $W = W_0 + \Delta_W$ , where  $\Delta_W \in \mathbb{R}^{m \times d}$  denotes the parameter change induced by post-training.

Let  $x \in \mathbb{R}^d$  denote the input hidden state to this matrix for an old example. The original layer output is  $z_0 = W_0 x$ , while the post-trained layer output is  $z = W x = W_0 x + \Delta_W x$ .

Thus, the change on the old hidden state  $x$  is

$$z - z_0 = \Delta_W x. \quad (3)$$

The local functional drift can be therefore measured as  $\|\Delta_W x\|^2$ . If this quantity is zero, then the output drift will also be zero. So, we first replace the data replay regularization from the final output level to the hidden state level:

$$\mathcal{L} = \sum_{i=1}^n \mathcal{L}(x_i, y_i) + \sum_{k=1}^m \sum_{j=n+1}^{n+u} \|\Delta_{W_k} h_{k_j}\|, \quad (4)$$

where  $n$  and  $u$  are the sizes of the new task dataset  $\mathcal{D}_{\text{new}}$  and the replay dataset  $\mathcal{D}_{\text{replay}}$ . We note that  $h$  denote the hidden state of a token rather than a sequence.  $m$  is the number of trainable parameter matrices, determined by the model and training method (e.g., if we are finetuning on a small dataset, we may choose to train only a small set of parameters) while independent of dataset sizes. For the conciseness of presentation, we will omit the subscript  $k$  hereafter. Nevertheless, we acknowledge the limitation that the cost grows with  $m$  and will discuss this issue in §??. Therefore, we rewrite Eq. 4 in the following more concise form:

$$\mathcal{L} = \sum_{i=1}^n \mathcal{L}(x_i, y_i) + \lambda \sum_{j=n+1}^{n+u} \|\Delta_W h_j\|. \quad (5)$$

In conclusion, we will use Eq. 5 as the objective to retain the old data behavior. The core motivation behind it is that we in fact do not need to learn  $\mathcal{D}_{\text{old}}$  as new knowledge, but rather we only need to keep the behavior unchanged.

The next question is that the regularization term  $\sum_u \|\Delta_W h_j\|$  is still linearly related to the dataset size  $u$ , and it needs to be re-computed at every gradient descent step. To address this issue,

**OneReplayer.** To avoid repeatedly scan the old dataset, we try to transform  $\{h_j\}_u$  into a fixed size matrix (independent of  $u$ ), so as to reduce the cost of reusing it.

For a collection of old hidden states  $H_{\text{old}} = [h_1, \dots, h_u] \in \mathbb{R}^{d \times u}$ , the total squared drift is

$$\|\Delta_W H_{\text{old}}\|_F^2 = \sum_{j=1}^u \|\Delta_W h_j\|_2^2 \quad (6)$$

Defining the empirical second moment matrix

$$C = \frac{1}{u} H_{\text{old}} H_{\text{old}}^\top = \frac{1}{u} \sum_{j=1}^u h_j h_j^\top, \quad (7)$$

we obtain

$$\frac{1}{u} \|\Delta_W H_{\text{old}}\|_F^2 = \text{tr}(\Delta_W C \Delta_W^\top), \quad (8)$$

where  $\text{tr}(\cdot)$  denotes the trace of a matrix.

Equivalently, if  $h \sim \mathcal{D}_{\text{old}}$  denotes a hidden state sampled from the old activation distribution, then

$$C = \mathbb{E}_h [h h^\top] \quad (9)$$

and

$$\mathbb{E}_h [\|\Delta_W h\|_2^2] = \text{tr}(\Delta_W C \Delta_W^\top) \quad (10)$$

This identity is the basis of OneReplayer. It shows that the average functional change of a linear layer on old hidden states can be computed from a fixed matrix  $C$ , without revisiting individual old examples during each gradient descent step. Under this formulation, OneReplayer aims to approximate the functional preservation benefit of replay while replacing example-level replay with fixed matrix memories.

As a result, Eq. 5 is equal to

$$\mathcal{L} = \sum_{i=1}^n \mathcal{L}(x_i, y_i) + \lambda \text{tr}(\Delta_W C \Delta_W^\top). \quad (11)$$

In the following sections, we denote  $R = \text{tr}(\Delta_W C \Delta_W^\top)$ .

#### 4.1 COMPUTATIONAL COMPLEXITY

This section will give a formal computational complexity comparison between our method and the vanilla replay method (i.e., Eq. 2). Our method introduces the additional computation of  $\Delta_W C \Delta_W^\top$  during training, whereas vanilla replay incurs additional computation by processing extra replay

| Method         | New Val Loss          | IFEVAL |       | MultiIF |       |
|----------------|-----------------------|--------|-------|---------|-------|
|                |                       | PSA    | ISA   | PLA     | ILA   |
| Base           | 12.82                 | 66.54  | 74.70 | 49.58   | 73.34 |
| FT             | $6.04 \times 10^{-2}$ | 55.08  | 65.23 | 31.47   | 60.65 |
| EWC            | $9.43 \times 10^{-2}$ | 65.99  | 73.62 | 48.66   | 72.43 |
| Vanilla Replay | $7.93 \times 10^{-2}$ | 65.43  | 73.74 |         |       |
| OneReplay      | $6.17 \times 10^{-2}$ | 66.17  | 73.38 |         |       |

Table 1: The performance of different finetuning methods on both the new task and instruction following dataset.

samples. From Eq. 4, we know that the computational cost decreases as the number of tunable parameters decreases. Therefore, we consider full-parameter fine-tuning, where all model parameters are tunable, to characterize the upper bound on the computational cost of our method.

**Settings.**  $h \in \mathbb{R}^{d_{in}}$  is the hidden state of a token,  $W \in \mathbb{R}^{d_{out} \times d_{in}}$  is the original weight matrix, and  $\Delta_W \in \mathbb{R}^{d_{out} \times d_{in}}$  is the weight update.  $C \in \mathbb{R}^{d_{in} \times d_{in}}$  is the pre-stored covariance matrix.

Evaluating the replay regularizer  $R = \text{tr}(\Delta_W C \Delta_W^\top)$  requires two matrix multiplications. The first one is  $\Delta_W C$ :

$$\Delta_W C = \Delta_W [c_1, c_2, \dots, c_{d_{in}}], \quad (12)$$

where  $c_i \in \mathbb{R}^{d_{in}}$  is a column of  $C$ , and it is equivalent to processing a sequence of  $d_{in}$  tokens. We denote  $\Delta_W C = W_c \in \mathbb{R}^{d_{out} \times d_{in}}$ . Similarly,  $W_c \Delta_W^\top$  is equivalent to processing  $d_{out}$  tokens.

So, the computational of  $R$  is equivalent to that of processing  $d_{in} + d_{out}$  tokens. This means that we can **make use of the whole replay dataset at every training step** with only  $d_{in} + d_{out}$  additional tokens, with  $d_{in} + d_{out}$  being a fixed constant irrelevant to the dataset size and the batch size. That’s to say, the cost of OneReplay is lower than that of vanilla replay (i.e., Eq. 2) as long as the sampled replay mini-batch contains more than  $d_{in} + d_{out}$  tokens, regardless of the fact that vanilla replay uses only a sampled mini-batch to approximate the replay dataset at each training step, while  $R$  contains the whole replay dataset. This means that

Note that in practice, the cost of OneReplay can be even less. First, compared to vanilla replay,  $R$  only involves multiplications associated with the model’s weight matrices and does not require computing pairwise attention scores among tokens, thereby avoiding a substantial amount of attention-related computation. Second, in parameter-efficient fine-tuning settings where only a small subset of parameters is updated (e.g., fine-tuning selected layers or using LoRA), the overhead becomes even smaller. This is because the computation of  $R$  depends only on the weight updates  $\Delta_W$  rather than the original weights  $W$ . Consequently, when the number of tunable parameters is only a small fraction of the total model parameters, the additional computational overhead introduced by  $R$  is nearly negligible. In such settings, the computational overhead of replay becomes negligible, making OneReplay comparable even to fine-tuning without replay.

## 5 EXPERIMENTS

**Baselines.** We consider four controlled baselines. Base: the base unfinetuned model. FT: finetuning without replay (i.e., only use the new task dataset). Vanilla Replay: finetuning with vanilla replay (i.e., Eq. 2). OneReplay: our method (i.e., Eq. 11).

## 6 SUBMISSION OF CONFERENCE PAPERS TO ICLR 2025

ICLR requires electronic submissions, processed by <https://openreview.net/>. See ICLR’s website for more instructions.

If your paper is ultimately accepted, the statement `\iclrfinalcopy` should be inserted to adjust the format to the camera ready requirements.

The format for the submissions is a variant of the NeurIPS format. Please read carefully the instructions below, and follow them faithfully.

## 6.1 STYLE

Papers to be submitted to ICLR 2025 must be prepared according to the instructions presented here.

Authors are required to use the ICLR L<sup>A</sup>T<sub>E</sub>X style files obtainable at the ICLR website. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for rejection.

## 6.2 RETRIEVAL OF STYLE FILES

The style files for ICLR and other conference information are available online at:

<http://www.iclr.cc/>

The file `iclr2025_conference.pdf` contains these instructions and illustrates the various formatting requirements your ICLR paper must satisfy. Submissions must be made using L<sup>A</sup>T<sub>E</sub>X and the style files `iclr2025_conference.sty` and `iclr2025_conference.bst` (to be used with L<sup>A</sup>T<sub>E</sub>X2<sub>ε</sub>). The file `iclr2025_conference.tex` may be used as a “shell” for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in sections 7, 8, and 9 below.

## 7 GENERAL FORMATTING INSTRUCTIONS

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing of 11 points. Times New Roman is the preferred typeface throughout. Paragraphs are separated by 1/2 line space, with no indentation.

Paper title is 17 point, in small caps and left-aligned. All pages should start at 1 inch (6 picas) from the top of the page.

Authors’ names are set in boldface, and each name is placed above its corresponding address. The lead author’s name is to be listed first, and the co-authors’ names are set to follow. Authors sharing the same address can be on the same line.

Please pay special attention to the instructions in section 9 regarding figures, tables, acknowledgments, and references.

There will be a strict upper limit of 10 pages for the main text of the initial submission, with unlimited additional pages for citations.

## 8 HEADINGS: FIRST LEVEL

First level headings are in small caps, flush left and in point size 12. One line space before the first level heading and 1/2 line space after the first level heading.

### 8.1 HEADINGS: SECOND LEVEL

Second level headings are in small caps, flush left and in point size 10. One line space before the second level heading and 1/2 line space after the second level heading.

#### 8.1.1 HEADINGS: THIRD LEVEL

Third level headings are in small caps, flush left and in point size 10. One line space before the third level heading and 1/2 line space after the third level heading.

## 9 CITATIONS, FIGURES, TABLES, REFERENCES

These instructions apply to everyone, regardless of the formatter being used.

### 9.1 CITATIONS WITHIN THE TEXT

Citations within the text should be based on the `natbib` package and include the authors' last names and year (with the "et al." construct for more than two authors). When the authors or the publication are included in the sentence, the citation should not be in parenthesis using `\citet{}` (as in "See Hinton et al. (2006) for more information.>"). Otherwise, the citation should be in parenthesis using `\citep{}` (as in "Deep learning shows promise to make progress towards AI (Bengio & LeCun, 2007).>").

The corresponding references are to be listed in alphabetical order of authors, in the REFERENCES section. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

### 9.2 FOOTNOTES

Indicate footnotes with a number<sup>1</sup> in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).<sup>2</sup>

### 9.3 FIGURES

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction; art work should not be hand-drawn. The figure number and caption always appear after the figure. Place one line space before the figure caption, and one line space after the figure. The figure caption is lower case (except for first word and proper nouns); figures are numbered consecutively.

Make sure the figure caption does not get separated from the figure. Leave sufficient space to avoid splitting the figure and figure caption.

You may use color figures. However, it is best for the figure captions and the paper body to make sense if the paper is printed either in black/white or in color.

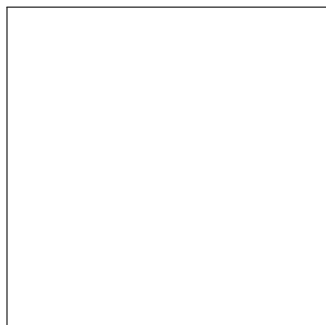


Figure 1: Sample figure caption.

### 9.4 TABLES

All tables must be centered, neat, clean and legible. Do not use hand-drawn tables. The table number and title always appear before the table. See Table 2.

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<sup>1</sup>Sample of the first footnote

<sup>2</sup>Sample of the second footnote

Table 2: Sample table title

| PART     | DESCRIPTION                       |
|----------|-----------------------------------|
| Dendrite | Input terminal                    |
| Axon     | Output terminal                   |
| Soma     | Cell body (contains cell nucleus) |

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

## 10 DEFAULT NOTATION

In an attempt to encourage standardized notation, we have included the notation file from the textbook, *Deep Learning* Goodfellow et al. (2016) available at [https://github.com/goodfeli/dlbook\\_notation/](https://github.com/goodfeli/dlbook_notation/). Use of this style is not required and can be disabled by commenting out `math_commands.tex`.

### Numbers and Arrays

|                           |                                                                                |
|---------------------------|--------------------------------------------------------------------------------|
| $a$                       | A scalar (integer or real)                                                     |
| $\mathbf{a}$              | A vector                                                                       |
| $\mathbf{A}$              | A matrix                                                                       |
| $\mathbf{A}$              | A tensor                                                                       |
| $\mathbf{I}_n$            | Identity matrix with $n$ rows and $n$ columns                                  |
| $\mathbf{I}$              | Identity matrix with dimensionality implied by context                         |
| $\mathbf{e}^{(i)}$        | Standard basis vector $[0, \dots, 0, 1, 0, \dots, 0]$ with a 1 at position $i$ |
| $\text{diag}(\mathbf{a})$ | A square, diagonal matrix with diagonal entries given by $\mathbf{a}$          |
| $\mathbf{a}$              | A scalar random variable                                                       |
| $\mathbf{a}$              | A vector-valued random variable                                                |
| $\mathbf{A}$              | A matrix-valued random variable                                                |

### Sets and Graphs

|                                         |                                                                                                     |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|
| $\mathbb{A}$                            | A set                                                                                               |
| $\mathbb{R}$                            | The set of real numbers                                                                             |
| $\{0, 1\}$                              | The set containing 0 and 1                                                                          |
| $\{0, 1, \dots, n\}$                    | The set of all integers between 0 and $n$                                                           |
| $[a, b]$                                | The real interval including $a$ and $b$                                                             |
| $(a, b]$                                | The real interval excluding $a$ but including $b$                                                   |
| $\mathbb{A} \setminus \mathbb{B}$       | Set subtraction, i.e., the set containing the elements of $\mathbb{A}$ that are not in $\mathbb{B}$ |
| $\mathcal{G}$                           | A graph                                                                                             |
| $\text{Pa}_{\mathcal{G}}(\mathbf{x}_i)$ | The parents of $\mathbf{x}_i$ in $\mathcal{G}$                                                      |

### Indexing

|     |                                                                              |                                                                                                             |
|-----|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 378 | $a_i$                                                                        | Element $i$ of vector $\mathbf{a}$ , with indexing starting at 1                                            |
| 379 | $\mathbf{a}_{-i}$                                                            | All elements of vector $\mathbf{a}$ except for element $i$                                                  |
| 380 | $A_{i,j}$                                                                    | Element $i, j$ of matrix $\mathbf{A}$                                                                       |
| 381 | $\mathbf{A}_{i,:}$                                                           | Row $i$ of matrix $\mathbf{A}$                                                                              |
| 382 | $\mathbf{A}_{:,i}$                                                           | Column $i$ of matrix $\mathbf{A}$                                                                           |
| 383 | $\mathbf{A}_{i,j,k}$                                                         | Element $(i, j, k)$ of a 3-D tensor $\mathbf{A}$                                                            |
| 384 | $\mathbf{A}_{:,:,i}$                                                         | 2-D slice of a 3-D tensor                                                                                   |
| 385 | $\mathbf{a}_i$                                                               | Element $i$ of the random vector $\mathbf{a}$                                                               |
| 386 | <b>Calculus</b>                                                              |                                                                                                             |
| 387 | $\frac{dy}{dx}$                                                              | Derivative of $y$ with respect to $x$                                                                       |
| 388 | $\frac{\partial y}{\partial x}$                                              | Partial derivative of $y$ with respect to $x$                                                               |
| 389 | $\nabla_{\mathbf{x}} y$                                                      | Gradient of $y$ with respect to $\mathbf{x}$                                                                |
| 390 | $\nabla_{\mathbf{X}} y$                                                      | Matrix derivatives of $y$ with respect to $\mathbf{X}$                                                      |
| 391 | $\nabla_{\mathbf{X}} y$                                                      | Tensor containing derivatives of $y$ with respect to $\mathbf{X}$                                           |
| 392 | $\frac{\partial f}{\partial \mathbf{x}}$                                     | Jacobian matrix $\mathbf{J} \in \mathbb{R}^{m \times n}$ of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$     |
| 393 | $\nabla_{\mathbf{x}}^2 f(\mathbf{x})$ or $\mathbf{H}(f)(\mathbf{x})$         | The Hessian matrix of $f$ at input point $\mathbf{x}$                                                       |
| 394 | $\int f(\mathbf{x}) d\mathbf{x}$                                             | Definite integral over the entire domain of $\mathbf{x}$                                                    |
| 395 | $\int_{\mathbb{S}} f(\mathbf{x}) d\mathbf{x}$                                | Definite integral with respect to $\mathbf{x}$ over the set $\mathbb{S}$                                    |
| 396 | <b>Probability and Information Theory</b>                                    |                                                                                                             |
| 397 | $P(\mathbf{a})$                                                              | A probability distribution over a discrete variable                                                         |
| 398 | $p(\mathbf{a})$                                                              | A probability distribution over a continuous variable, or over a variable whose type has not been specified |
| 399 | $\mathbf{a} \sim P$                                                          | Random variable $\mathbf{a}$ has distribution $P$                                                           |
| 400 | $\mathbb{E}_{\mathbf{x} \sim P}[f(\mathbf{x})]$ or $\mathbb{E}f(\mathbf{x})$ | Expectation of $f(\mathbf{x})$ with respect to $P(\mathbf{x})$                                              |
| 401 | $\text{Var}(f(\mathbf{x}))$                                                  | Variance of $f(\mathbf{x})$ under $P(\mathbf{x})$                                                           |
| 402 | $\text{Cov}(f(\mathbf{x}), g(\mathbf{x}))$                                   | Covariance of $f(\mathbf{x})$ and $g(\mathbf{x})$ under $P(\mathbf{x})$                                     |
| 403 | $H(\mathbf{x})$                                                              | Shannon entropy of the random variable $\mathbf{x}$                                                         |
| 404 | $D_{\text{KL}}(P \  Q)$                                                      | Kullback-Leibler divergence of $P$ and $Q$                                                                  |
| 405 | $\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$             | Gaussian distribution over $\mathbf{x}$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$   |
| 406 | <b>Functions</b>                                                             |                                                                                                             |
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|-----|-----------------------------------------|------------------------------------------------------------------|
| 432 | $f : \mathbb{A} \rightarrow \mathbb{B}$ | The function $f$ with domain $\mathbb{A}$ and range $\mathbb{B}$ |
| 433 | $f \circ g$                             | Composition of the functions $f$ and $g$                         |
| 434 | $f(x; \theta)$                          | A function of $x$ parametrized by $\theta$ . (Sometimes we write |
| 435 |                                         | $f(x)$ and omit the argument $\theta$ to lighten notation)       |
| 436 | $\log x$                                | Natural logarithm of $x$                                         |
| 437 | $\sigma(x)$                             | Logistic sigmoid, $\frac{1}{1 + \exp(-x)}$                       |
| 438 | $\zeta(x)$                              | Softplus, $\log(1 + \exp(x))$                                    |
| 439 | $\ \mathbf{x}\ _p$                      | $L^p$ norm of $\mathbf{x}$                                       |
| 440 | $\ \mathbf{x}\ $                        | $L^2$ norm of $\mathbf{x}$                                       |
| 441 | $x^+$                                   | Positive part of $x$ , i.e., $\max(0, x)$                        |
| 442 | $\mathbf{1}_{\text{condition}}$         | is 1 if the condition is true, 0 otherwise                       |

## 11 FINAL INSTRUCTIONS

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the REFERENCES section; see below). Please note that pages should be numbered.

## 12 PREPARING POSTSCRIPT OR PDF FILES

Please prepare PostScript or PDF files with paper size “US Letter”, and not, for example, “A4”. The `-t letter` option on `dvips` will produce US Letter files.

Consider directly generating PDF files using `pdflatex` (especially if you are a MiKTeX user). PDF figures must be substituted for EPS figures, however.

Otherwise, please generate your PostScript and PDF files with the following commands:

```
dvips mypaper.dvi -t letter -Ppdf -G0 -o mypaper.ps
ps2pdf mypaper.ps mypaper.pdf
```

### 12.1 MARGINS IN LATEX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below using `.eps` graphics

```
\usepackage[dvips]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.eps}
```

or

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

for `.pdf` graphics. See section 4.4 in the graphics bundle documentation (<http://www.ctan.org/tex-archive/macros/latex/required/graphics/grfguide.ps>)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command.

## AUTHOR CONTRIBUTIONS

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

## ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

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## A APPENDIX

You may include other additional sections here.